

MH1101 Tutorial 10 (Week 11)

Reference: Topic 5.3, 5.4 (Lecture Notes)

1. Determine whether the series is absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1}$

(b) $\sum_{n=0}^{\infty} \frac{(-3)^n}{(2n+1)!}$

(c) $\sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n} \right)^n$

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n e^{1/n}}{n^3}$

(e) $\sum_{n=1}^{\infty} \frac{n5^{2n}}{10^{n+1}}$

2. For which positive integers k is the following series convergent?

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}.$$

3. Prove that $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$ for all x .

Hint: Use the Ratio Test.

4. Prove the Root Test:

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence and assume that the following limit exists:

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

- (i) If $L < 1$, then $\sum a_n$ converges absolutely.
- (ii) If $L > 1$ or $L = \infty$, then $\sum a_n$ diverges.
- (iii) If $L = 1$, the Root Test is inconclusive. The series may converge or diverge.

5. Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms, and let $r_n = \frac{a_{n+1}}{a_n}$. Suppose $\lim_{n \rightarrow \infty} r_n = L < 1$.

Let $R_n = a_{n+1} + a_{n+2} + \cdots$. If $\{r_n\}$ is a decreasing sequence and $r_{n+1} < 1$, show that

$$R_n \leq \frac{a_{n+1}}{1 - r_{n+1}}.$$

Answer Keys.

1. (a) converges conditionally (b) converges absolutely (c) converges absolutely (d) converges absolutely (e) diverges 2. diverges for $k = 1$, converges (absolutely) for $k \geq 2$